

Grid Tied Solar PV Cost Proposal



August 2025
1 Bolt Ave | 366 kWp

Clear
& Simple

About Us

Emergent Energy

At **Emergent Energy**, we believe that energy investments should be **clear and simple**. Guided by our core values of **authenticity, integrity, and accountability**, we cut through complexity to help businesses make informed energy decisions with confidence. Our mission is to simplify energy investments by providing expert insights and tailored quality solutions that drive growth, empower people, and promote sustainability. With a dynamic and collaborative approach, we transform the way businesses engage with energy—turning challenges into opportunities for innovation and lasting impact.

4 Offices Nationwide	55MWp Under Performance Management	390+ Completed Solar Sites
14 Years In Activity	>800 Million ZAR of in-house funding	120MWp Installed To Date

With Fedgroup as our shareholders we have in house access to funds for our energy solutions. And being part of the **Fedgrop Platform of Companies**, we're able to provide holistic solutions to property owners and businesses alike that covers solar asset management and utilities monitoring with Matriarch and Techsitter and an ISP with Intellicomms.

The Brief

Client:

Company Name	Annenberg Property Group
Point Of Contact	Andries Louw
Title	Director

Your activity:

Industry	Commercial Cold Storage
Working Days	Monday to Sunday
Working Hours	24 hours
Annual Shutdown Days	3 Days

Your Property

Address	1 Bolt Avenue, Montague Gardens
Erf Number	ERF 4687
Roof Covering	Kliplok
Usable Roof Area (sqm)	Entire roof Area Usable

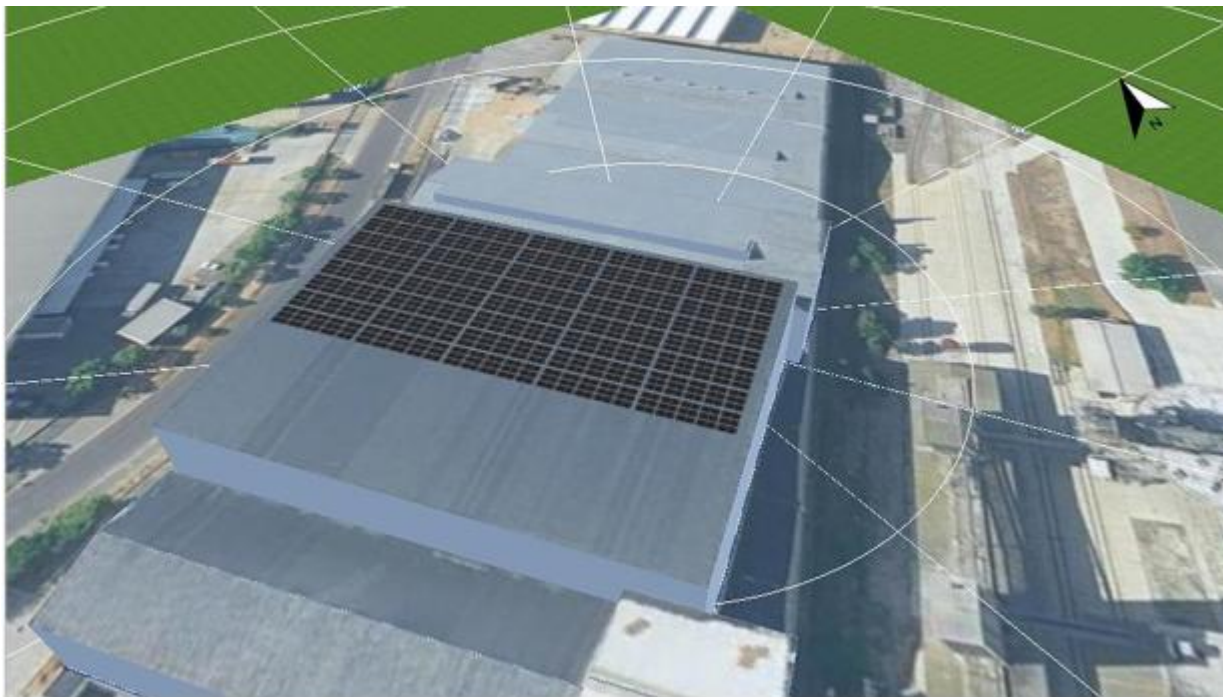
Electrical load information:

Electricity Supplier	ESKOM
Tariff Description	Miniflex TOU
Maximum Demand (or CB Size)	1000 kVA
Energy Consumption (monthly average)	220 000 kWh
Load Data Provided	Metering Data and bills
Backup System	Diesel Generator

System Design

Using bespoke solar PV design software, the proposed location has been modelled and optimised, bearing in mind variables such as available space, panel orientation, shading patterns, and obstructions.

The design is used to calculate the solar PV system's energy yield i.e. the expected amount of electricity that can be produced throughout the year. We use both our historical experience and accurate modelling to ensure that the proposed system design will meet your needs.



The above design could be altered at a later stage to accommodate construction related constraints

Roof considerations

Roof selected	1 North East-facing roof
Roof sheet type	Kliplok
Inclination	5°
Major obstructions	none
Current Blended Tariff	R2,46
Your Designer	DEAN

System design highlights

Sizing Rationale	Consumption
System Size	366 kWp
AC Capacity	300 kVa
Solar Annual kWh	560 838 kWh
Current Blended Tariff	R2,46
Electrical Savings (Y1)	R1 378 654

Component Selection

Emergent Energy ensures that all suppliers meet the highest quality and workmanship standards. Equipment costs typically account for around 80% of the total project costs, the majority of which are made up of PV panel and inverter costs. Emergent Energy therefore sources and selects suppliers based on the following criteria to ensure a long-term return on investment:

- Quality
- Price
- Local presence and support
- Financial strength (to ensure that warranties have a real value)

Our team has selected the following components to match your site's requirements.

Component	Proposed Supplier	Value Added / Warranty
Solar PV Modules	Jinko Solar 600 x 620Wp	12-year warranty + 25-year performance guarantee at 85% of nominal output
Solar Inverter	Huawei Inverters 3 x Sun2000-100KTL-M2	5-year warranty (extendable)
Mounting Structure	Schletter	10-year warranty
Monitoring System	Huawei Fusion Solar	Track the 2-year workmanship warranty and receive 3-months of performance reports
Consumption and Production Metering	Consumption – Accuenergy or Janitza (LV or MV)	Consumption monitoring and reverse power blocking.
	Production – Techsitter Revenue Meter	Revenue-grade metering for solar production.
Diesel Generator Integration	Not included	Enables your renewable energy plant to generate during power outages safely.
Additional	Cat-ladder, Steel Inverter Rack	

System Performance

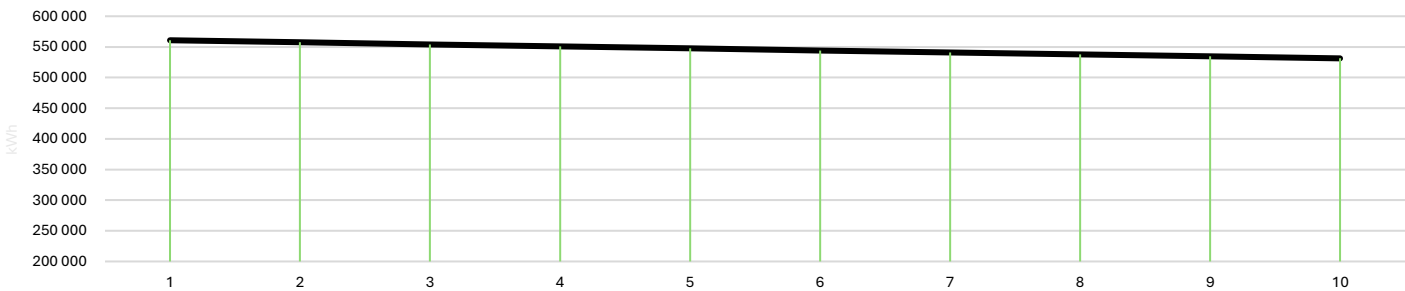
1. Performance drivers

Panels degradation factor	0,6%	The ratio by which the solar panel performance decreases yearly.
Solar reference yield: *	1572 kWh/kWp	Measures the efficiency of your Solar system.
Annual energy yield (Y1)	560 838 kWh	The amount of solar energy generated annually.
Blended solar tariff	R2,46/kWh	Reviewed tariff based on utility and feed-in tariffs.
Solar savings in Year1	R1 378 654	Energy Yield x Solar Tariff.

**While our software makes use of a 15-year weather dataset, the actual yield can vary from the values above.*

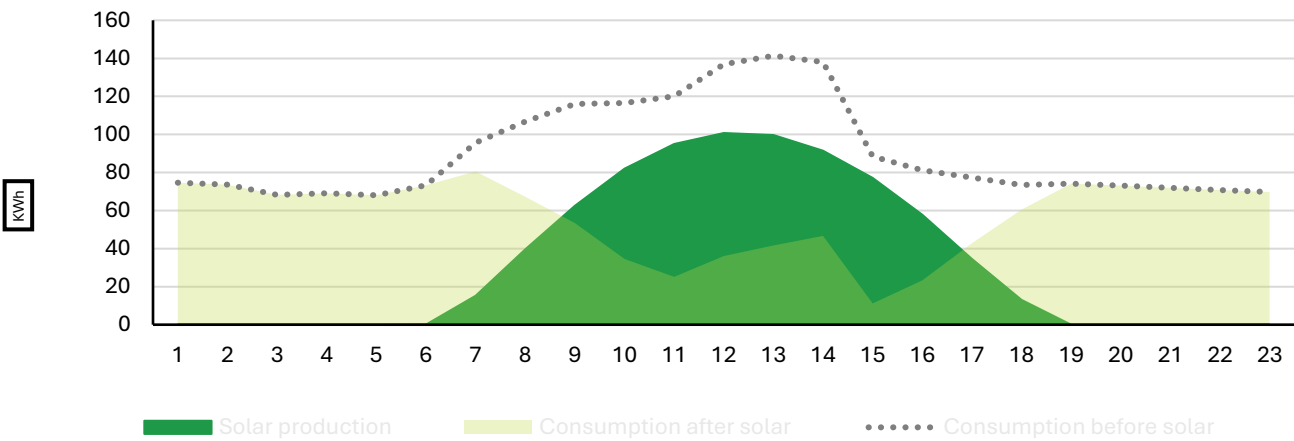
**The above data and results have been extracted from the information provided by the client.*

2. Annual production (kWh) over 20 years



**A solar panel has two warranties: performance and equipment guarantee. A solar panel's performance warranty will typically guarantee at least 90% of initial production after 10 years and 80% at 25 years. The equipment warranty covers defects or failures over 10-12 years.*

3. Your average daily energy profile



**The above graph is displayed for illustration purpose only and intends to showcase the assumptions made on the facility load profile and the solar offset. The graph depicts the peak performance of the system, which is dependent on weather conditions.*

Warranty & Services

1. The Workmanship Warranty

The Return on Investment of your solar system is achieved over time; therefore, appropriate warranties must be provided post-construction to safeguard the performance of your system.

Accordingly, following the completion of the solar system installation, a 2-year Workmanship Warranty will be activated. The scope is summarised below and detailed further in Appendix 4.

Service Summary	Workmanship Warranty	Performance Monitoring and Advisory	Customised Operation & Maintenance	Performance Guarantee
3 Months Performance Reporting	✓			
Automated Reporting		✓	✓	✓
Telephonic support and Remote Troubleshooting	✓	✓	✓	✓
Dashboard Access	Extra	✓	✓	✓
Continuous Performance Monitoring	-	✓	✓	✓
Performance Reporting	-	✓	✓	✓
Annual Energy Report	-	✓	✓	✓
Detailed Annual Inspection	-	✓	✓	✓
Annual Visual On Site Inspection	-	-	As per plan	✓
Preventative Maintenance Tasks	-	-	As per plan	✓
Cleaning of solar modules	-	-	As per plan	✓
Any Corrective Maintenance	-	-	-	✓
On-site Troubleshooting and labour for fault repairs	-	-	-	✓

The above range of services, including the administration and management of the Workmanship Warranty period, is performed by Matriarch Asset Managers (Pty) Ltd an independently operating subsidiary of Emergent Energy. Matriarch will appoint a dedicated Asset Manager to oversee the operation of your solar system during this period.

Because of the independent operation of the Asset Manager from Emergent Energy's project delivery team during the warranty period, their mandate dictates to safeguard performance, identify defects and thereby ensure favourable returns for the asset owner.

2. Performance Reporting

Optimising the performance of Solar PV systems is essential in terms of increased revenue generation for the asset owner.

Following the successful completion of the project and as part of the Workmanship Warranty, the Client will receive 3 monthly Performance Reports. Should the Client wish to extend the Performance Reporting period and enter a Performance Monitoring contract, please reach out to your Dedicated Asset Manager.

Remote Performance Monitoring and Performance Reporting track key performance metrics, report on maintenance tasks and indicate potential performance improvements.

By comparing the actual performance of the Solar PV system to what was initially modelled, Matriarch's Performance Report gives you a concise view of your system performance and solar revenue.

3. Cloud-based Remote System & Performance Monitoring

We use the internet to remotely monitor the site performance and gather data. The data is then stored and processed on our cloud-based platform to be translated into diagnostics and predictive analytics.

4. Inverter Monitoring Platform Access

You will be given access to the inverter manufacturers' monitoring platform. Through the cell phone application, you can track live performance data. Please enquire directly with Dedicated Asset Manager to book a basic training session.

5. Telephonic Technical Support

For technical support post-installation, our Asset Managers remain available to address any queries you may have over the telephone.

6. Yearly Site Visit

The solar system will be inspected by our team once a year to ensure the system is safe and sound for the duration of the workmanship warranty.

Financials

To better define the business case, we use our internally developed Energy Audit and Energy Model tool to calculate your annual yield and energy production. These inputs assist us in determining your cash flow over the next 20 years. We then model your current tariff against projected electricity price increases and compare it to the savings you would receive after investing in the solar PV system.

Please refer to Appendix 3 for further information on the financial drivers and assumptions.

Self-funded

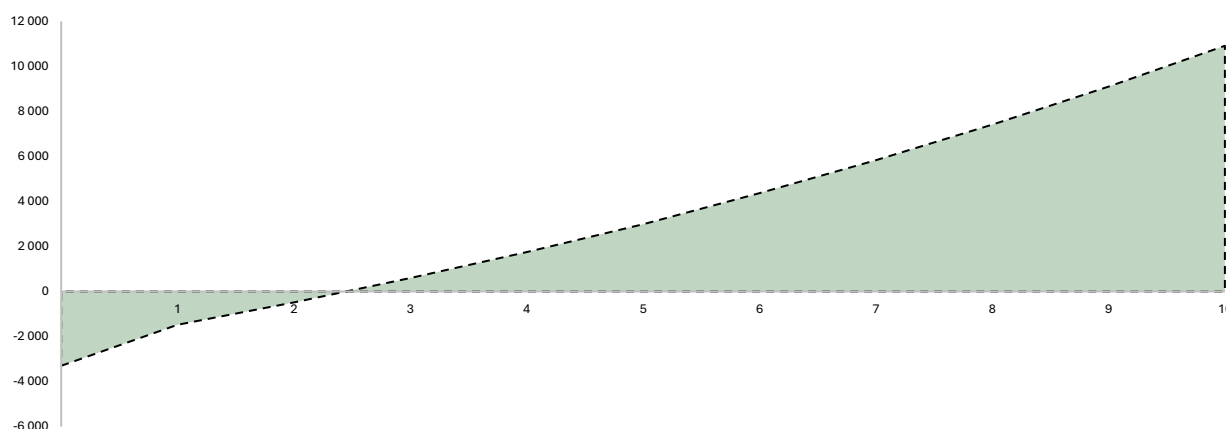
CAPEX	R3 283 811
IRR	40%
Payback	2 Years
Cost per Wp	R8,97

Self-Funded

the best asset class for steady & safe returns

Financial inputs (1000)	Year	1	2	3	4	5	6	7	8	9	10	11-20
Capital Expenditure	-3 284											
Tax Deductions		0	0	0								
Electricity Savings		1 379	1 480	1 589	1 706	1 831	1 966	2 110	2 265	2 432	2 611	39 371
Grid Losses		-28	-30	-32	-34	-37	0	0	0	0	0	0
Asset Management Budget		-131	-141	-150	-161	-172	-184	-197	-211	-226	-241	-3 570
Yearly Net Cashflow	-3 284	1 082	1 162	1 248	1 340	1 439	1 585	1 702	1 828	1 963	2 108	31 864
Cumulated Cashflow	-3 284	-2 202	-1 040	208	1 548	2 987	4 572	6 274	8 102	10 065	12 173	44 037

Cumulated Cash Flow



40 %

IRR

R3 283 811

Total Outlay

2 years

Payback (years)

R8,97

Cost per Wp

Project Delivery

Emergent Energy has a streamlined project management approach with an efficient organisational structure and dynamic process. Our project management team is carefully structured to serve you through all aspects of the project implementation.

Our Pillars



Our Project Delivery Team



Project Manager

Project managers review proposals and plans and address any concerns throughout the project duration. From site visits to the commissioning and handover stage. PM's oversee the project preparation, acceptance, commissioning and handover. The project manager handles all technical matters, selects suppliers, and leads your installation team.



Project Engineer

The project engineer develops the construction plans based on site inspections and discussions with suppliers and installation partners.



Head Of Project

The Head of Project provides support to the project team by coordinating project plans, liaising with key stakeholders, and ensuring that projects are delivered in a timely manner.



Chief Operations Officer

Our COO is the backstop to our projects team, ensuring all projects are delivered to standards and our clients are kept informed throughout the project delivery phase.



Installation Partners (IP)

These are either the Emergent Energy internal team or contractors selected by Emergent Energy.

Performance Reporting

The “Red Team” is an independent quality surveying department that surveys the construction of solar sites and ensures that quality standards are enforced and achieved by the project delivery teams.

It is required for the solar PV system to stick to a range of strict guidelines, regulations, and safety standards. The audit led by our team consists of 100+ checkpoints and aims at ensuring that the solar asset is built to last and perform in accordance with national standards, guidelines, and international best practices.

The service is designed to identify, record, and locate the primary source of anomalies. The site survey covers the below items:

- Installation and compliance
- Inverter mechanical and electrical inspection and compliance
- DC Stringing, cable runs, earthing and safety
- Module and Mounting Structure
- Module Inspection for Damage and Soiling
- Compliance to SANS 10142 -1 wiring code and NRS
- Electronic communication functionality and installation



Next Steps

Emergent Energy prides itself on acting proactively and transparently. We, therefore, ensure a clear and succinct communication flow between our teams and the client throughout the project.

Clients are encouraged to liaise with their relationship manager or project manager whenever they wish to. The table below breaks down the next steps from the acceptance of the proposal to the project completion.

Note that the funded projects need to go through a due diligence process set by the funder for the funds to be released and the below to be actioned. Funding approval lead times may vary according to the funder, please reach out to your relationship managers should you need clarity or support.

Planning Phase: Contract Signature to Kick-off

1)	Proposal Acceptance/Signature
24/48h	If you wish to proceed and accept this proposal, please refer to Appendix 1. By signing this proposal acceptance form, you are accepting our offer in its entirety. Please make sure that you've indicated the option you have decided on.
2a)	Deposit Invoice
5 working days	Following receipt of the proposal acceptance form, Emergent Energy's financial department will be in touch to invoice the agreed deposit amount. Confirmation of receipt of the deposit amount enables our team to start the procurement process. Please refer to Appendix 2&3 for further information on the payment terms and conditions.
2b)	SSEG and Structural Sign-Off
	Concurrently with the above, our teams will liaise with your utility provider to register the solar system and send an independent structural engineer to ensure that the roof is structurally sound. Although the risks of SSEG and of structural failures are low, Emergent Energy manages the risks on the client's behalf and will keep you informed about delays and remedial work needs if any.
3)	Internal kick-off
20 working days	Once the construction plan has been defined and material procured, the project execution and timeline are reviewed internally before being presented to the client for approval.
4)	Client kick-off
5 working days	Your relationship manager will introduce you to our project delivery team. The action plan and project milestones are then presented to you. No project starts until the plan is fully cleared and agreed upon by the client. Topics such as panel layout, inverter placements, general logistics and health and safety matters are discussed.

Next Steps

Construction Phase: Site Establishment to Handover

5)	Site Establishment and Start of Construction
	Once the project plan has been reviewed with our installation team, the project is ready to start. Tenant Packs can be shared so that the client can better communicate with your tenant or the users of the property. During the entire duration of the project, the client will be able to liaise directly with the project and relationship manager. Additionally, updates on the status of the site are sent to the client weekly.
6)	Practical Completion
	All components are installed. This is the physical completion of work. The Red Team surveys the site to address snags and ensures that the quality of permanent items is standard. This enables the system to be energised and the testing to start.
7)	Testing and Commissioning
	After energising the solar system, our engineers test and monitor the performance of the system and ensure that all components operate as they should. The last snags are addressed, and the project is wrapped up, ready for the client handover.
8)	Client Handover
	The project is now completed. Our team will physically hand over the project and unpack the deliverables. The status of the SSEG process, minor tests and snags remaining will be addressed during the meeting.
9)	Handover to the Asset Managers
	The solar system is live and performing, the client will be introduced to his Dedicated Asset Manager overseeing the operation of the solar system and compiling the monthly performance reports.

Appendices

Appendix 1: Proposal Acceptance Form

I, Andries Louw

("the client representative"),

acting as Director (role within client's organisation)

on behalf of Golden Lakes Investments (Pty) Ltd ("the client), and duly authorised to sign, hereby accept the Cost Proposal reference: **260825_1 Bolt Ave_366kWp_CP** and terms as detailed herein.

I am intending to proceed with the following option:

☒

Self-Funded

Signature:



Date:

1 September 2025

Milestone	Milestone Assessment Guide	Percentage	Amount Due
Acceptance and Initial Deposit	Signature Of This Proposal	35%	R1 149 333,69
Delivery of Panels	Receipt of Panels on site	30%	R985 143,17
Delivery of inverter	Receipt of inverters on site	20%	R656 762,11
Practical Completion	All provisional test successful finalized	10%	R328 381,06
Handover	All provisional test successful finalized	5%	R164 190,53
Total		100%	R3 283 810,55

Appendix 2: Important planning information and general conditions

1. General Conditions

1.1 Proposal duration - this proposal is valid for a period of 10 days from the date of submission to the date of deposit received. The selection of major equipment is based on foreign exchange rates, stock availability and lead time at the submission date and any delays in receiving the initial deposit will require a reassessment of the pricing, project timelines and the selected suppliers for major equipment.

1.2 Component Selection - we may propose changes to main components, specifically solar panels, and inverters, with similar specifications to that offered depending on stock availability at the time of signing, subject to your approval.

1.3 An independent structural engineer will verify that the roof substructure can withstand the weight of the solar PV system. Emergent Energy always conducts a first analysis in the pre-feasibility study, but this verification can only be completed after acceptance of this proposal. Should the structural engineer recommend modifications to the roof substructure to accept the solar panels, all costs associated with the modifications are excluded from this proposal. If the modification costs are determined to change the business case and the project is cancelled, any costs incurred to date by Emergent are to be reimbursed by the client.

1.4 Internet connection - unless otherwise agreed between parties, a stable internet connection is to be provided on-site by the client before the commissioning of the system. This internet connection allows for the remote monitoring of the system.

1.5 Insurance – Emergent Energy is responsible for the insurance of the components for the duration of the installation up until the Project Handover, after which all insurance risk passes onto the client's insurance.

1.6 Batteries, Diesel Generators and Generator Integration – the selection of the size of rating of any backup technologies are completed using all the information supplied by the Client.

The operational limitations and ultimate performance of the generator are subject to several factors including cooling systems, step-changes of your load or start-up times.

The warranty and conditions associated with the supply and installation of any diesel generator are supplied separately from the solar system proposal.

Where Integration of your solar system with any diesel generator is included, Emergent Energy cannot warrant the operating levels of the solar system, its penetration to the generator loads or any diesel saving.

2: All options excl. PPA and RRI:

2.1 Ownership - ownership and warranties of the panels, inverters, mounting structures and all other items are transferred to the client once the components have been delivered to the site and all payments received in line with the payment milestones detailed in Appendix 3.

2.2 Solar yield projections- the solar system performance projections while based on empirical weather data, are theoretical and subject to external variables. Emergent Energy completes these yield projections with the utmost accuracy, using the weather data, solar modelling software, your working hours or hourly usage profiles. However, the actual yields may differ from what is presented. Emergent Energy will install quality equipment and commission a quality system as well as include all possible variables in the yield estimates, but Emergent Energy cannot be held liable should the actual yield differ from the projections. This includes yield or energy lost to grid limitation or curtailment due to changes in working hours or any reduction in energy usage affecting solar production. Please refer to The Brief or the Specific Yield for these assumptions.

Initial here: _____



2.3 Utility Grid Connection, Small Scale Embedded Generation (SSEG) Approval and NERSA Registration.

As part of the overall process for legally connecting a solar system to an existing electricity supply and network, additional fees or costs could be charged by the Utility provider (Eskom or the local municipality) directly to the Client for the grid connection of the solar system. Emergent Energy can offer management of these connection costs separately from this proposal.

Further, in South Africa, it is required by law that all electricity consumers wishing to install an SSEG solar system, regardless of generation capacity, register their system with the relevant administration. Emergent Energy offers to act on the client's behalf free of charge to facilitate this SSEG application process and includes the required sign-off by a professional engineer and other supporting documentation required for this application and ultimate approval. We act proactively to manage the process to avoid delays but cannot be held liable for any delays that may occur during the application process for connection or for grid feed-in approval.

Finally, the Electricity Regulations Act requires all electricity generators to register their system with the National Energy Regulator of South Africa (NERSA). Again, Emergent Energy will facilitate this process on your behalf once all the required documentation is in hand but cannot be held liable for any delays in receiving the acknowledgement of registration from NERSA.

connection point or DB board at your premises.

Any additional costs for connecting the system to non-compliant connections will be at the client's expense.

It is assumed that the current electricity supply is compliant with current standards and legislation and that installation have followed best industry practices.

Note that if your electrical network is using 3-phase power, the solar PV system will equally feed the three phases assuming your network

is balanced across the 3 phases. Any unbalanced loads or high single-phase loads could lead to energy curtailment across the phases and energy loss.

Initial here: _____



Appendix 3: Financial model considerations and payment terms

1: General

1.1 Financial models are only projections genuinely aiming for accuracy, but actual numbers can differ from what is presented.

1.2 Electricity price increases are assumed to increase by 8% for years 1-20.

1.3 The maintenance fee is modelled at between 1-2% of the system cost and is an allowance only

1.4 The CPI Inflation rate is modelled at 5%

1.5 The bank funded cashflow is modelled at an interest rate of, 11%.

1.6 Carbon Tax: A solar PV system will generate tax exemption. The assumption made on a tax at R120/t of carbon, growing by inflation rate.

1.7 Degradation factor: 0,6% yearly as per manufacturer's recommendation,

1.8 All figures are **exclusive of VAT**.

2: Exchange rate

1 USD = 17,70 ZAR

2.1 Exchange rate assumptions: This proposal and the associated costing are linked to Foreign Exchange rates. The costs exposed to exchange rate fluctuations are typically between 50 – 60% of the total system cost. While we attempt to absorb some of the exchange rate variability through our long-term partnerships with suppliers, hedging and Forward Cover contracts, it may be the case that final pricing will need to be adjusted at the time of receiving the deposit payment.

3: Solar Blended Tariff and Electricity tariff

The Solar Blended Tariff is calculated from your current or past electricity bills and the tariff charged by your Utility provider. All solar energy produced is valued at this Solar Blended Tariff, which is a function of the potential solar production and the amount of this production that is consumed and amount of that is fed-in to the grid or curtailed (lost).

INSERT UTILITY YEAR (Solar Blended Tariff: R2,46/kWh

4: Depreciation

As of the 01/03/2025 100%-year 1 solar asset depreciation (based on a 27% tax rate) for all system sizes.

5: Payment terms:

The Payment Milestones in Appendix 1 are designed to minimise the Client's exposure to risk and balance the cashflow for both the client and Emergent. Payments are due within 7 days of invoice and Emergent Energy reserves the right to charge interest for late payments and to remove any installed equipment it deems necessary to cover costs incurred for unpaid milestones without limiting Its rights to further recourse.

Initial here:  _____

Appendix 4: Workmanship Warranty

Matriarch Asset Managers has been appointed by Emergent Energy to carry out the services related to the 2-year (24 months) warranty on installation and workmanship.

The Workmanship Warranty starts once the project has been commissioned and handed over to the client.

All Fault classifications are provided at the sole discretion of the Matriarch Asset manager.

Fault Classifications:

- Category 1 Wear and Tear faults are classified as failures over time occurring as a result of the normal course of use as intended by the relevant design criteria. Examples are fuse failures, aging internet cable failures, UPS failures.
- Category 2 Out-of-normal faults are defined as failures occurring outside of the normal course of use or failures occurring as a result, either indirectly or directly, from events not considered in the original design criteria. Examples are loadshedding-related failures, mechanical or malicious damage including theft and vandalism, hail damage.
- Category 3 Defects are classified as failures resulting from inherent shortcomings in the design, installation, manufacture or supply of labour, components or materials that ought to have been considered, and as such would be covered under the terms and conditions of the warranty provided by the supplier, while within the warranty period. Examples are communication board failures, un-torqued mounting structure, loose connections.

For a Category 3 fault resulting in a warranty claim, we will motivate and justify the claim to successfully conclude the process, however we cannot guarantee that all claims will be approved. Any and all on-site labour to remove or replace components under warranty is excluded and will be priced separately under a Bolt-On service. Please ask your project developer for more information relating to this.

Matriarch Asset Managers does not directly provide any warranties to the client for any labour or components. All component warranties are provided by the OEM or Component supplier.

This warranty shall not apply if the components installed have been damaged by unreasonable use, negligence, service, or modification performed by anyone other than Emergent Energy and Matriarch Asset Managers or by any other cause unrelated to the failure of materials and quality of workmanship. Repairs of these faults are considered Additional Services.

Fees for Additional Services covered during the Workmanship Warranty:

- R560 per hour for Remote or Telephonic Assistance
- R680 per hour for On Sit Assistance by Technician
- R1250 per hour for On Sit Assistance by Engineer or Project Manager
- R5800 per day for On Site repairs for sub-contractor team including Health and Safety requirements.

All the other materials, consumables, incidentals, equipment hire and travel cost will be reimbursed at the cost plus 8% fee.

Please note that Workmanship Warranty could be voided if anyone other than Emergent Energy or Matriarch Asset Managers performs works on the solar PV system or its components without prior approval from Matriarch Asset Managers.



Appendix 5: Company Information

1: Company structure

Emergent Energy is an independent solar photovoltaic company and is 100% owned by Fedgroup. Fedgroup are in turn independently owned and managed and committed to remaining that way. Their independence buys them the ability to do things differently and better. As a result, we are able to offer the best financing solutions in the African market for projects.

We take time to understand the role that solar investment will play for you, and we bring clarity to the process by taking care of the details and sharing what matters. In doing so, we empower you to make the right decisions for your businesses to grow.

2: Board Of Director



Robert Timmis / Chief Executive Officer

Robert is a professionally registered engineer with extensive experience in the energy sector of South Africa, having worked in nuclear, conventional and renewable power generation. His focus is on the commercial aspect of this highly technical field to bring a better understanding of energy investments to businesses across South Africa. Rob leads Emergent as its Chief Executive.



Grant Field / Chief Director

Grant has been the CEO of Fedgroup since 2015 and boasts more than a decade's experience in the Beneficiary Trust industry, which forms the cornerstone of care on which the business operates. He achieved his BSc in Engineering at the University of the Witwatersrand and has since completed his PDBA, MBA and various industry-related regulatory exams.



Dayne Sage / Technical Director

Dayne is a professionally registered electrical engineer and installation electrician. This combination of practical and theoretical knowledge, combined with 10 years in the Solar PV industry, allows him to keep an eye on all the operational and technical aspects of Emergent Energy's business. He also has previously worked in the vehicle manufacturing industry and enjoys bringing some lean manufacturing learnings into the engineering of the business.



Johannes / Chief Operating Officer

With an Industrial Engineering degree from Stellenbosch University, Johannes brings a wealth of experience in operations management, process improvement, and strategic implementation to his role as Chief Operating Officer at Emergent Energy. Having held key leadership positions across various industries, Johannes has developed a strong track record in optimizing organizational efficiency, driving business growth, and leading high-performing teams.